# Stress-Testing Instruction Chatbots for Economics Experiments:
## Which Workflow and Model Best Preserve Treatment Integrity?

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Preliminary — June 1, 2026

An instruction chatbot deployed inside an economics experiment is not a help tool—it is part of the *manipulation*. It must clarify rules and refuse to supply advice, expected-value computations, or strategic hints. We built an adversarial **stress-test harness** that pits synthetic "subject" personas against the bot over thousands of multi-turn conversations and scores every reply with an independent LLM judge on three metrics: bias-leak (the integrity metric), factual error, and incoherence. We then asked a single deployment question: *which pipeline architecture and which model leak the least?*

**Design.** A two-phase plan over two scenarios (an easier mid-experiment context and a $\sim 2\times$ harder cold-start context) and seven adversarial personas, 15 runs $\times$ 15 turns per cell. **Phase A** (15,750 judged turns) screens five architectures, from a single all-in-one call to a classifier $\rightarrow$ router $\rightarrow$ per-class corrector, all at the cheapest model. **Phase B** escalates the winner one lever at a time: a corrector pass, reasoning effort (low/medium/high), and model capability.

**Headline results.**

- **Simpler is safer.** The single-call pipeline is Pareto-dominant on bias-leak, cost, and latency in every cell. Every added agent (classifier, low-confidence probe gate, corrector) leaves bias the same or worse—a 1.2–4.1$\times$ penalty—because the extra steps reframe and re-surface the subject's question, which is itself a contamination channel.
- **Capability is the only lever that works.** Holding the winning architecture fixed, upgrading the base model (`nano`$\rightarrow$`mini`) cut bias-leak **46%** on the easier scenario and **68%** on the harder one, improved factuality, and ran faster at non-binding cost. Reasoning effort was neutral-to-harmful at every level tested; a corrector pass made bias *worse*.
- **Integrity is a per-conversation property.** Single-turn leak risk is small but compounds across a 15-turn dialogue; we report per-conversation consistency, not single-shot accuracy.

**Caveat.** The headline upgrade cell currently shares one model between bot and judge, so a small self-leniency artefact cannot yet be excluded; an independent-judge confirmation cell is in flight. We therefore state the finding as *"model capability is the lever that works,"* not as a final production recommendation.

**Full results, interactive figures, and methods:**
`can-celebi.github.io/chatbot-for-experiments`